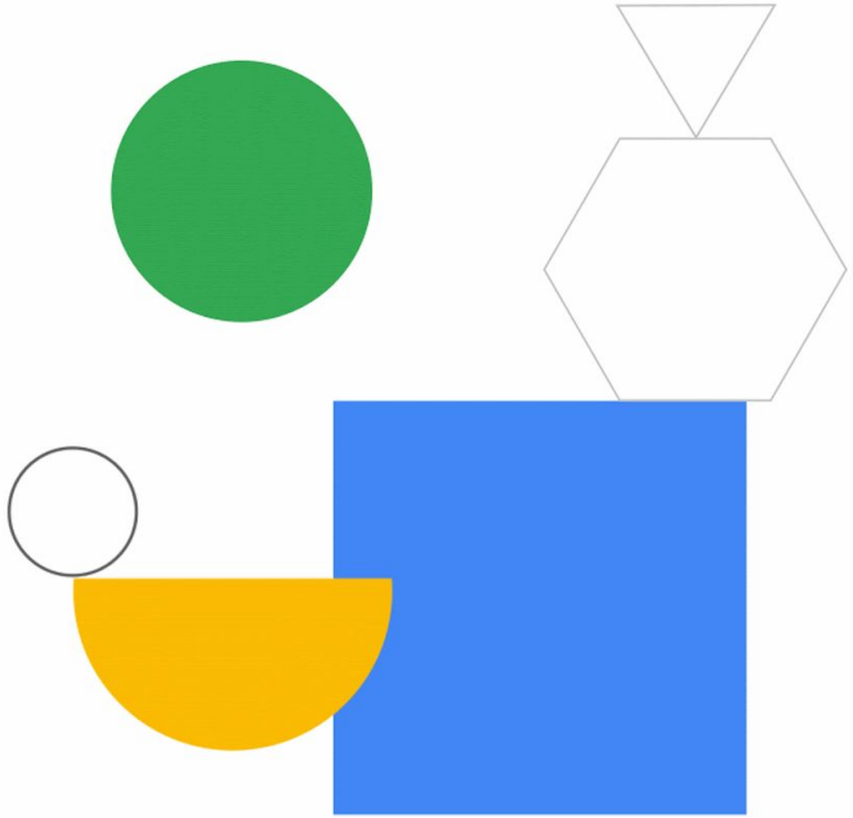
Google Cloud

Terms and Concepts



Objectives

Upon completion of this module, you will be able to:

01 Explain the Terraform workflow.

02 Create basic configuration files within Terraform.

03 Explain the purpose of a few Terraform commands.

04 Describe the Terraform Validator tool.

05 Create, update, and destroy Google Cloud resources using Terraform.

Topics

01 The Author phase

02 Terraform commands

03 Terraform Validator Tool

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Terraform workflow

1	2	3	4	5
Scope	**Author**	**Initialize**	**Plan**	**Apply**
Confirm the resources required for a project.	Author the configuration files based on the scope.	Download the provider plugins, and initialize the module.	View execution plan for resources created, modified, or destroyed.	Create actual infrastructure resources.

Terraform Workflow

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Terraform directory

- Terraform uses configuration files to declare an infrastructure element.
- The configuration is written in terraform language with a .tf extension.
- A configuration consists of:
  - A root module/ root configuration
  - Zero or more child modules
  - Variable.tf (optional but recommended)
  - Outputs.tf (optional but recommended)
- Terraform commands are run on the working directory.

```
-- main.tf
   -- servers/
      -- main.tf
      -- providers.tf
      -- variables.tf
      -- outputs.tf
```

Root module

Child module

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HashiCorp Configuration Language (HCL)

- Terraform's configuration language for creating and managing API-based resources
- Configuration language, not a programming language.
- Includes limited set of primitives such as variables, resources, outputs and modules.
- Includes traditional statements or control loops.

Syntax

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

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HCL syntax

- Blocks
- Arguments
- Identifiers
- Expressions
- Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
  # A custom mode VPC network resource
   name                      = mynetwork
   auto_create_subnetworks   = false
}
```

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HCL syntax

- **Blocks**
- Arguments
- Identifiers
- Expressions
- Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
# custom mode network definition
   name                      = mynetwork
  auto_create_subnetworks    = false
}
```

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HCL syntax

Blocks

Arguments

Identifiers

Expressions

Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
# custom mode network definition
   name                        = mynetwork
  auto_create_subnetworks   = false
}
```

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HCL syntax

Blocks

Arguments

Identifiers

Expressions

Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
# custom mode network definition
  name                      = mynetwork
  auto_create_subnetworks   = false
}
```

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HCL syntax

Blocks

Arguments

Identifiers

Expressions

Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <VALUE/EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
# custom mode network definition
   name                      = mynetwork
   auto_create_subnetworks   = false
}
```

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HCL syntax

Blocks

Arguments

Identifiers

Expressions

Comments

```
<BLOCK TYPE> "<BLOCK LABEL>" "<BLOCK LABEL>" {
  # Block body
  <IDENTIFIER> = <EXPRESSION> #Argument
}
```

```
resource "google_compute_network" "default" {
   # custom mode network definition
   name                      = mynetwork
   auto_create_subnetworks   = false
}
```

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Resources

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- Resources are code blocks that define the infrastructure components.

  Example: Cloud Storage Bucket

- Terraform uses the `resource type` and the `resource name` to identify an infrastructure element.

```
resource "resource_type" "resource_name" {
  #Resource specific arguments
}
```

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Example for resources

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- The keyword `resource` is used to identify the block as the cloud infrastructure component.
- Resource type is `google_storage_bucket`.
- The resource name is `example-bucket`.

```
resource "google_storage_bucket" "example-bucket" {
  name          = "<unique-bucket-name>"
  location      = "US"
}
```

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Resource arguments

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- The arguments differ based on the resource type.
- Some arguments are required, others are optional.

```
resource "google_storage_bucket" "example-bucket" {
  name          = "<unique-bucket-name>" //Required
  location      = "US"
}

resource "google_compute_instance" "my_instance" {
  name         = "test"
  machine_type = "e2-medium"
  zone         = "us-central1-a"
  boot_disk {
    initialize_params {
      image = "debian-cloud/debian-9"
    }
  }
}
```

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Provider (1 of 2)

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- Terraform downloads the provider plugin in the root configuration when the provider is declared.
- Providers expose specific APIs as Terraform resources and manage their interactions.

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
      version = "4.23.0"
    }
  }
}

provider "google" {
  # Configuration options
   project = <project_id>
   region  = "us-central1"
}
```

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Provider (2 of 2)

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- Provider configurations belong in the root module of a Terraform configuration.
- Arguments such as project and region can be declared within the provider block.

```
terraform {
  required_providers {
    google = {
      source  = "hashicorp/google"
      version = "4.23.0"
    }
  }
}

provider "google" {
  # Configuration options
   project = <project_id>
   region  = "us-central1"
}
```

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Provider versions

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- The version argument is optional, but recommended.
- The version argument is used to constrain the provider to a specific version or a range of versions.

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
      version = "4.23.0"
    }
  }
}

provider "google" {
  # Configuration options
}
```

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Variables

- Parameterize resource arguments to eliminate hard coding its values

  Example: Region, project ID, zone, etc.
- Define a resource attribute at run time or centrally in a file with a .tfvars extension.

Storage bucket

main.tf	variables.tf	terraform.tfvars
Declare storage bucket	Parameterize resource attribute	Define values at run time or in a file

`location`	`variable location {`
	`..`
	`}`
		`location=''US''`
		`storage_class="STANDARD"`

`unique_name`		
`storage_class`		
`versioning`		

CLI

```
$terraform apply

location=''US''
storage_class="STANDARD"
```

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Outputs values

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
```

- Output values are stored in `outputs.tf` file.
- Output values expose values of resource attributes.

```
output "bucket_URL" {
  value = google_storage_bucket.mybucket.URL
}
```

```
#terraform apply
Google_storage_bucket.mybucket: Creating...
Google_storage_bucket.mybucket: Creating complete after 1s []

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

bucket_URL = “https://storage.googleapis.com/my-gallery/..
```

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State

```
-- main.tf
-- providers.tf
-- variables.tf
-- outputs.tf
-- terraform.tfvars
-- terraform.tfstate
```

- Terraform saves the state of resources it manages in a state file.
- The state file can be stored:
  - Locally (default)
  - Remotely in a shared location

You do not modify this file.


```
{
  "version": 4,
  "terraform_version": "1.0.11",
  "serial": 3,
  "lineage": "822c3d96-0500-29cd-68e3-13101f2846f0",
  "outputs": {
    "vm_name": {
      "value": "terraform-test",
      "type": "string"
    }
  },
  "resources": [
    {
      "mode": "managed",
      "type": "google_compute_instance",
      "name": "default",
      "provider":
"provider[\"registry.terraform.io/hashicorp/google\"]",
      "instances": [
        {
```

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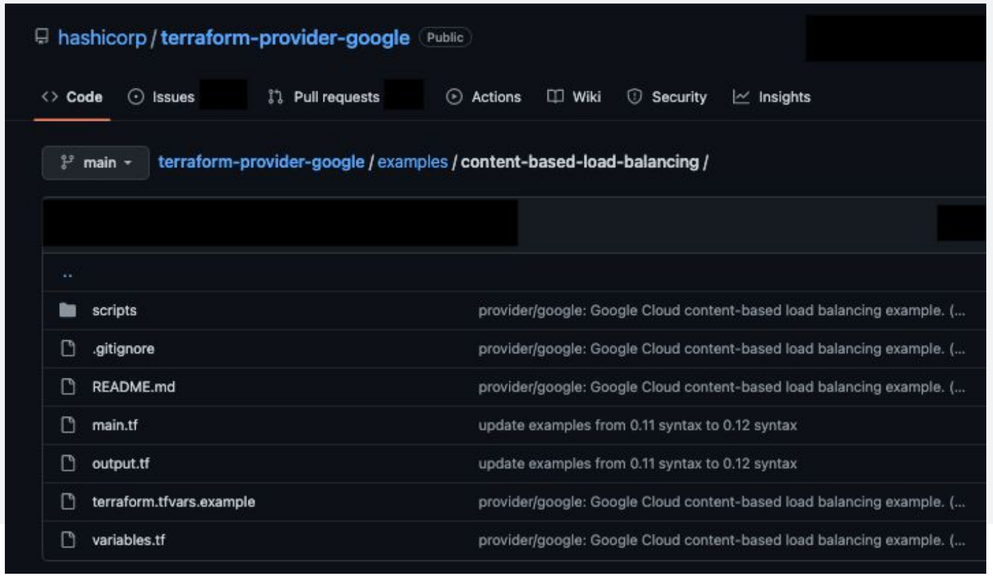
Modules

A Terraform module is a set of Terraform configuration files in a single directory.

- It is the primary method for code reuse in Terraform.
- There are 2 kinds of sources:
  - Local: Source within your directory
  - Remote: Source outside your directory.

```
-- instances/
   -- main.tf
   -- variables.tf
   -- outputs.tf
   -- terraform.tfvars
```

modules



Topics

01 The Author phase

02 Terraform commands

03 Terraform Validator tool

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Terraform commands

`terraform init`	Initialize the provider with plugin
`terraform plan`	Preview of resources that will be created after terraform apply
`terraform apply`	Create real infrastructure resources
`terraform destroy`	Destroy infrastructure resources
`terraform fmt`	Auto format to match canonical conventions

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Initialize Terraform using `terraform init`

1	2	3	4	5
Scope	**Author**	**Initialize**	**Plan**	**Apply**
Confirm the resources required for a project.	Author the configuration files based on the scope.	Download the provider plugins, and initialize the module.	View execution plan for resources created, modified, or destroyed.	Create actual infrastructure resources.

Terraform Workflow

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Initialize Phase:

`terraform init` downloads the provider plugins

```
-- main.tf
    -- servers/
       -- main.tf
       -- variables.tf
       -- outputs.tf
```

```
terraform {
  required_providers {
    google = {
      source = "hashicorp/google"
    }
  }
}
```

```
$ terraform init
Initializing the backend...

Initializing provider plugins...
- Finding latest version of hashicorp/google...
- Installing hashicorp/google v4.21.0...
- Installed hashicorp/google v4.21.0 (signed by HashiCorp)
..
Terraform has been successfully initialized!
```

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Preview resource action using `terraform plan`

1	2	3	4	5
Scope	**Author**	**Initialize**	**Plan**	**Apply**
Confirm the resources required for a project.	Author the configuration files based on the scope.	Download the provider plugins, and initialize the module.	View execution plan for resources created, modified, or destroyed.	Create actual infrastructure resources.

Terraform Workflow

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Plan Phase:

`terraform plan` creates an execution plan

```
resource "google_storage_bucket" "example-bucket"{
  name          = "student0313ab04569a94"
  location      = "US"
}
```

```
$terraform plan
Terraform will perform the following actions:
  # google_storage_bucket.example-bucket will be created
  + resource "google_storage_bucket" "example-bucket" {
      + force_destroy               = false
      + id                          = (known after apply)
      + location                    = "US"
      + name                        = "student0313ab04569a94"
      + storage_class               = "STANDARD"
      + uniform_bucket_level_access = (known after apply)
    }
Plan: 1 to add, 0 to change, 0 to destroy.
```

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Executes the actions proposed in a Terraform plan using `terraform apply`

1	2	3	4	5
Scope	**Author**	**Initialize**	**Plan**	**Apply**
Confirm the resources required for a project.	Author the configuration files based on the scope.	Download the provider plugins, and initialize the module.	View execution plan for resources created, modified, or destroyed.	Create actual infrastructure resources.

Terraform Workflow

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Appy Phase

`terraform apply` executes the plan

```
resource "google_storage_bucket" "example-bucket"{
  name     = "student0313ab04569a94"
  location = "US"
}
```

```
$terraform apply
Terraform will perform the following actions:
  # google_storage_bucket.example-bucket will be created
  + resource "google_storage_bucket" "example-bucket" {
      + force_destroy               = false
      + id                          = (known after apply)
      + location                    = "US"
      + name                        = "student0313ab04569a94"
      …
Apply changes: yes
google_storage_bucket.example-bucket: Creating...
google_storage_bucket.example-bucket: Creation complete after 1s [id=student0313ab04569a94]
Apply complete! Resources: 1 added, 0 changed, 0 destroyed.
```

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Code conventions

Formatting best practices:

- Separate meta arguments from the other arguments.
- Use two spaces for indentation.
- Align values at the equal sign.
- Place nested blocks below arguments.
- Separate blocks by one blank line.

[Style conventions]

```
resource “google_storge_bucket” “mybucket” {
   boot_disk { #nested arguments above
       initialize_params {
         image = "debian-cloud/debian-9"
       }
     }
   count = 2 #meta-argument in between
   machine_type=e2_micro #unaligned equal signs
   ..
}
```

```
resource “google_storge_bucket” “mybucket” {
   count          = 2 #meta-argument last

   machine_type   = e2_micro #align equal signs

   boot_disk { #nested arguments below
       initialize_params {
         image = "debian-cloud/debian-9"
       }
     }..
}
```

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terraform fmt

Terraform *fmt* can enforce code convention best practices.

Before `terraform fmt`:

```
resource “google_storge_bucket” “mybucket” {
    boot_disk { #nested arguments above
        initialize_params {
          image="debian-cloud/debian-9"
        }
    }
    count = 2 #meta-argument in between
    machine_type=e2_micro #unaligned equal signs
    ..
```

After `terraform fmt`:

```
resource “google_storge_bucket” “mybucket” {
    count            = 2 #meta-argument first
    machine_type     = e2_micro #align equal signs
    #line space before a nested block

    boot_disk { #nested arguments below
        initialize_params {
          image = "debian-cloud/debian-9"
        }
      }..
```

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terraform destroy command

```
$ terraform destroy
google_storage_bucket.example-bucket: Refreshing state... [id=student0313ab04569a94]
..
Terraform will perform the following actions:
  # google_storage_bucket.example-bucket will be destroyed
  - resource "google_storage_bucket" "example-bucket" {
      - force_destroy               = false -> null
      - id                          = "student0313ab04569a94" -> null
      - location                    = "US" -> null
      - name                        = "student0313ab04569a94" -> null
      ..
    }
Plan: 0 to add, 0 to change, 1 to destroy.
..
google_storage_bucket.example-bucket: Destroying... [id=student0313ab04569a94]
google_storage_bucket.example-bucket: Destruction complete after 0s
…
Destroy complete! Resources: 1 destroyed.
```

The **terraform destroy** command will destroy all the resources and its associated data from the main directory.

Topics

01 The Author phase

02 Terraform commands

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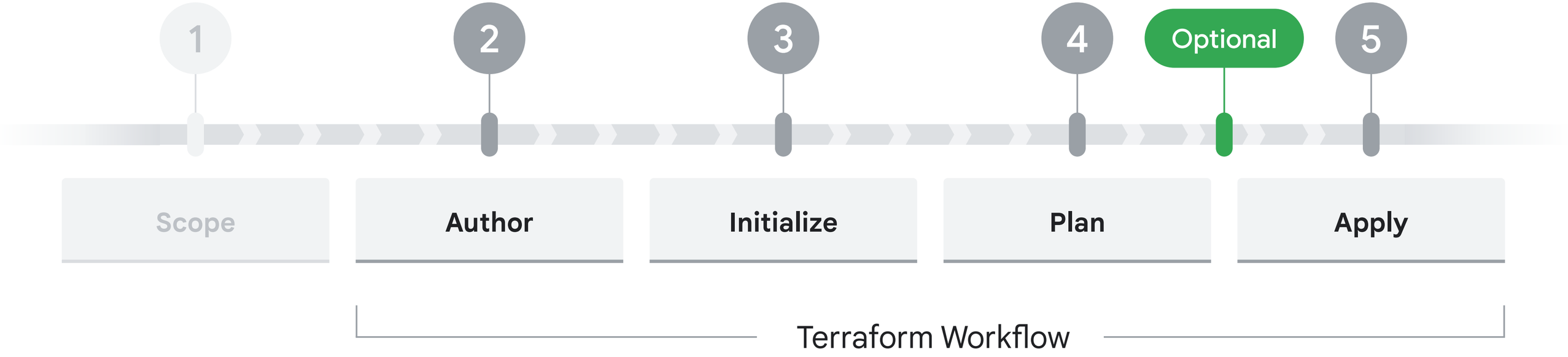
The validate phase

Validate

Runs pre-deployment checks against organizations policies.



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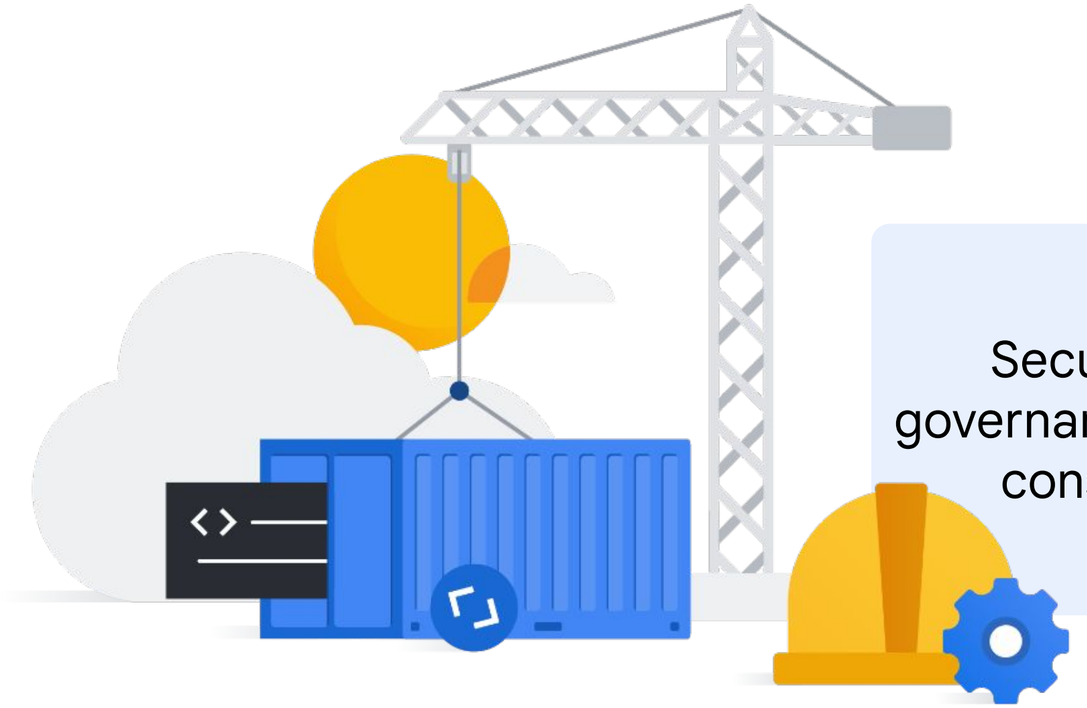
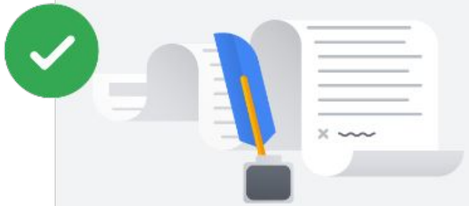
Why use the Terraform validator tool?



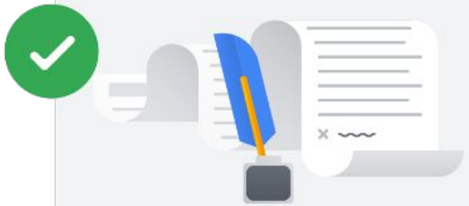
Security and governance team set constraints.

Automate validation

Governance policies



Security violation



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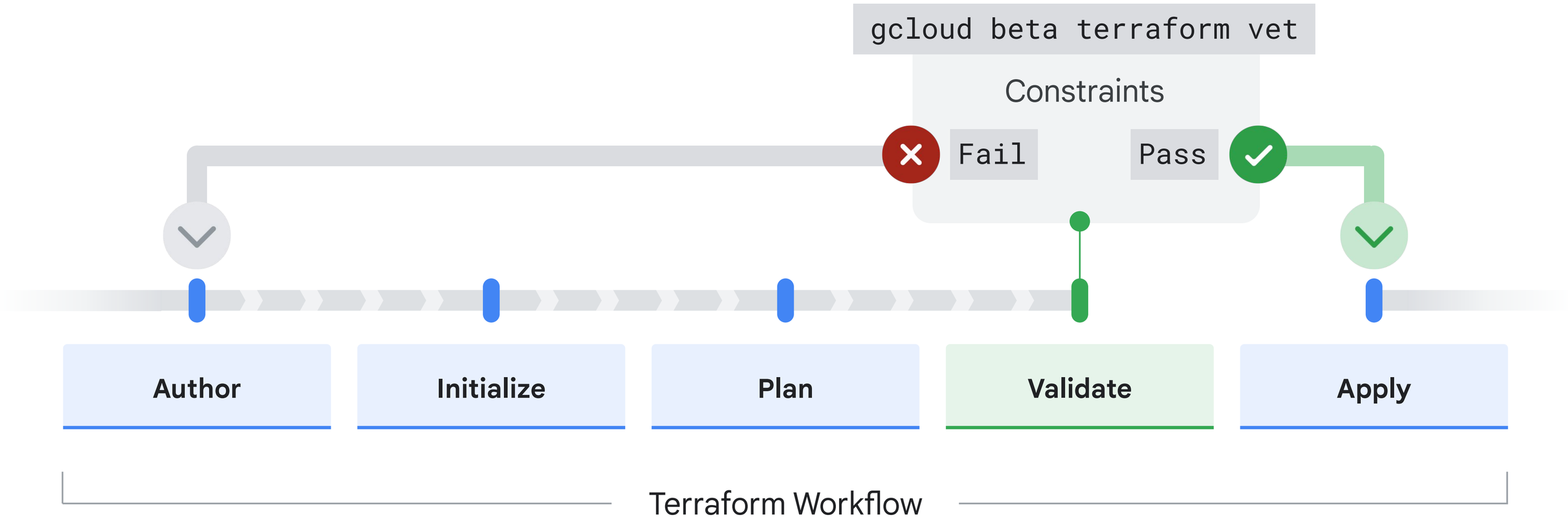
gcloud beta terraform vet



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Terraform Validator uses

01

Platform teams can add guardrails to infrastructure CI/CD pipelines to ensure all changes are validated.

02

Application teams and developers can validate Terraform configuration with organization's central policy library.

03

Security teams can create a centralized policy library to identify and prevent policy violations.

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Demo

Demonstrate Terraform Workflow

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Lab

Infrastructure as Code with Terraform

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Quiz

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Quiz | Question 1

Question

In which phase of the Terraform workflow can you run pre-deployment checks against the policy library?

A. Plan

B. Scope

C. Validate

D Initialize

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Quiz | Question 1

Answer

In which phase of the Terraform workflow can you run pre-deployment checks against the policy library?

A. Plan

B. Scope

C. Validate

D Initialize

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Quiz | Question 2

Question

Which command creates infrastructure resources?

A. terraform apply

B. terraform plan

C. terraform fmt

D terraform init

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Quiz | Question 2

Answer

Which command creates infrastructure resources?

A. terraform apply

B. terraform plan

C. terraform fmt

D terraform init

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Quiz | Question 3

Question

In which phase of the Terraform workflow do you write configuration files based on the scope defined by your organization?

A. Author

B. Initialize

C. Plan

D Scope

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Quiz | Question 3

Answer

In which phase of the Terraform workflow do you write configuration files based on the scope defined by your organization?

A. Author

B. Initialize

C. Plan

D Scope

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Module Review

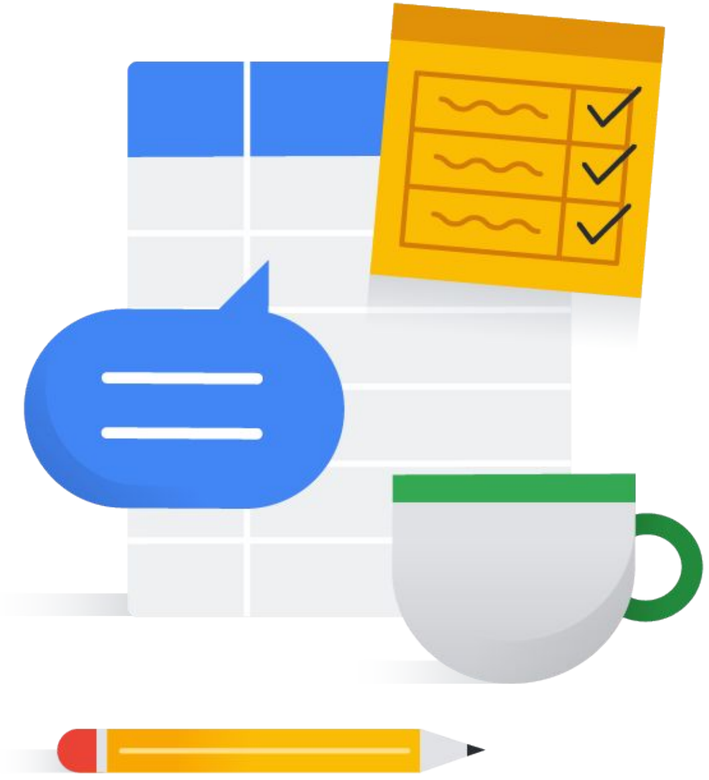
01 Explain the Terraform workflow.

02 Create basic configuration files within Terraform.

03 Explain the purpose of a few Terraform commands.

04 Describe the Terraform Validator tool.

05 Create, update, and destroy Google Cloud resources using Terraform.



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